DISCRETE STRUCTURES

CMSC250

Chapter 24: Cardinality and the Pigeonhole Principle

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Infinite sizes, diagonal arguments, and counting guarantees



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1 What does cardinality measure?

The *cardinality* $|A|$ of a finite set is its number of distinct elements. For infinite sets, we compare sizes using bijections: two sets have the same cardinality if a bijection exists between them.

Key idea. An infinite set can have the same cardinality as one of its proper subsets. This cannot happen for finite sets.

2 Countably infinite sets

A set is *countably infinite* if there is a bijection between it and $\mathbb{N} = \{0, 1, 2, \ldots\}$. A set is *countable* if it is finite or countably infinite. Informally, a countably infinite set can be listed as a sequence without omissions or repeats.

Worked example: even natural numbers

Question. Show that $E = \{0, 2, 4, 6, \ldots\}$ and $\mathbb{N}$ have equal cardinality.

Proof. Define $f : \mathbb{N} \to E$ by $f(n) = 2n$. If $f(a) = f(b)$ then $2a = 2b$, so $a = b$; hence f is injective. For any $e \in E$, the definition of an even nonnegative integer gives $e = 2n$ for some $n \in \mathbb{N}$, so $f(n) = e$; hence f is surjective. Therefore f is a bijection and $|E| = |\mathbb{N}|$. ■

$$\begin{array}{cccccccc} \mathbb{N} & 0 & 1 & 2 & 3 & 4 & \cdots \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ E & 0 & 2 & 4 & 6 & 8 & \cdots \end{array}$$

3 Why the integers are countable

Although the integers extend in both directions, they can be enumerated:

$$0, 1, -1, 2, -2, 3, -3, \ldots$$

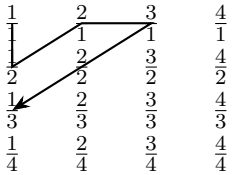
Each integer appears at a finite position. One explicit bijection $h : \mathbb{N} \to \mathbb{Z}$ is

$$h(n) = \begin{cases} n/2, & n \text{ even}, \\ -(n+1)/2, & n \text{ odd}. \end{cases}$$

The outputs begin $0, -1, 1, -2, 2, \ldots$; either ordering is valid.

4 Why rational numbers are countable

Every positive rational can be written as p/q with positive integers p, q. Place fractions in an infinite grid indexed by numerator and denominator and traverse the grid diagonally. Skip duplicate fractions (such as $1/2 = 2/4$). Every positive rational appears because its numerator and denominator lie on some finite diagonal. Include zero and negative rationals by interleaving signs. Therefore $\mathbb{Q}$ is countable.



A diagonal enumeration begins at the top-left

5 Uncountability and Cantor's diagonal argument

Not all infinite sets are countable. The real numbers in $(0, 1)$ are uncountable. The key idea is that *any proposed complete list* of real numbers in $(0, 1)$ misses a carefully constructed number. **Question.** Prove that $(0, 1)$ is uncountable.

Proof. Suppose, for contradiction, that all real numbers in $(0, 1)$ can be listed as $r_1, r_2, \ldots$. Choose decimal expansions not ending in infinitely many 9s. Construct $x = 0.d_1d_2d_3\ldots$ by setting $d_n = 1$ if the nth decimal digit of r_n is not 1, and $d_n = 2$ otherwise. Then each d_n is 1 or 2, so $x \in (0, 1)$ and no decimal ambiguity occurs. For every n, the nth digit of x differs from the nth digit of r_n. Thus $x \neq r_n$ for every n, contradicting the claim that the list contained every real in $(0, 1)$. Therefore $(0, 1)$ is uncountable. ■

Why the proof works. The constructed number differs from the first listed number in digit 1, the second in digit 2, and so on. Hence it cannot occur anywhere on the list.

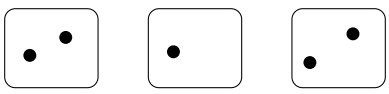
6 Power sets and Cantor's theorem

For a finite set with n elements, its power set has 2^n elements. More generally, Cantor's theorem states that $|A| < |\mathcal{P}(A)|$ for every set A, including infinite sets. **Question.** Explain why no function $f : A \to \mathcal{P}(A)$ can be surjective.

Proof. Let $f : A \to \mathcal{P}(A)$ be arbitrary. Define $D = \{a \in A : a \notin f(a)\}$. Suppose there is $d \in A$ with $f(d) = D$. Then $d \in D$ iff $d \notin f(d) = D$, a contradiction. Hence no element maps to D, and f is not surjective. ■

7 The pigeonhole principle

If more than n objects are placed into n boxes, at least one box contains at least two objects. The *generalized pigeonhole principle* says that placing N objects into k boxes forces some box to contain at least $\lceil N/k \rceil$ objects.



Five objects in three boxes force a repeated box

Worked example: birth months

Question. How many people guarantee at least two were born in the same month?

Solution. There are 12 months (boxes). With 13 people (objects), at least two share a birth month. Twelve people are insufficient for a guarantee because they could have distinct birth months.

Worked example: generalized principle

Question. Among 31 students, show that at least three were born in the same month.

Proof. If every month contained at most two of the 31 birthdays, the total would be at most $12 \cdot 2 = 24$, a contradiction. Therefore at least one month contains at least three birthdays. Equivalently, $\lceil 31/12 \rceil = 3$. ■

8 Pigeonhole proofs with remainders

Question. Prove that among any six integers, two have a difference divisible by five.

Proof. Every integer has exactly one remainder in $\{0, 1, 2, 3, 4\}$ when divided by 5. These are five possible boxes. Given six integers, two, say a, b, have the same remainder r. Thus $a = 5q + r$ and $b = 5t + r$ for integers q, t. Subtracting gives $a - b = 5(q - t)$, so $5 \mid (a - b)$. ■

9 Practice questions with solutions

Question. 1. Prove the set of odd natural numbers is countably infinite.

Proof. The map $f : \mathbb{N} \to \{1, 3, 5, \ldots\}$ given by $f(n) = 2n + 1$ is injective since equal outputs imply equal inputs, and surjective since every positive odd integer has this form. Therefore it is a bijection. ■**Question.** 2. How many socks must be selected to guarantee three socks of the same color if there are four colors?

Solution. At most two of each color can be chosen without obtaining three matching socks, for a total of 8. Therefore 9 guarantee three of one color (assuming enough socks exist). **Question.** 3. Among 11 integers, prove two have a difference divisible by ten.

Proof. There are ten possible remainders modulo ten. Eleven integers force two with the same remainder; subtracting their quotient–remainder representations shows their difference is a multiple of ten. ■**Question.** 4. Why does listing $0, 1, -1, 2, -2, \ldots$ establish countability of $\mathbb{Z}$?

Solution. Every integer appears at a finite position, with no repetitions. Therefore the list defines a bijection with the natural numbers.

10 Common mistakes

Do not equate countable with finite. A subset of a countable set is countable, but a subset of an uncountable set need not be uncountable. In pigeonhole arguments, state what the objects and boxes represent, and prove why the assignment to boxes is valid.